PATENT ANALYSIS REPORT

Project Title: Smart Plant Monitoring System

Report Title: Patent Analysis & Relevance Assessment

Prepared by: Kareliya Priyansh (2502031030003)

Soni Nishika (2502031030002)

Hirapara Nilakshi (2502031030047)

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1. Executive Summary

This report analyzes six patents related to Smart Plant Monitoring Systems, IoT-based agriculture control, and automated plant care devices. The objective is to evaluate technical overlap, assess infringement risk, and identify safe implementation strategies for our project.

Key Findings

- Multiple patents cover automated plant monitoring using sensors.
- IoT-based environmental control systems are widely patented.
- Smart containers and plant cultivation devices integrate monitoring features.
- Some patents focus on cloud and IoT integration.

Recommendation

To avoid infringement risk, the system should:

- Use multi-parameter adaptive logic
- Focus on small-scale home gardening
- Avoid direct replication of patented control methods
- Document unique architecture clearly

2. Methodology

1. Identified relevant patents from European Patent Office (EPO) and USPTO.
2. Reviewed patent title, filing date, and claims.
3. Compared independent claims with proposed project features.
4. Evaluated risk level (Low / Medium / High).

3. Patent Summary Table

No.	Patent ID	Title	Office	Date	Risk Level
1	EP3990913A1	Automated Plant Monitoring Systems and Methods	EPO	4 May 2022	Medium
2	US20240329614A1	Smart Plant Management System	USPTO	3 Oct 2024	Low
3	US10349584B2	System and Method for Plant Monitoring	USPTO	16 July 2019	Medium
4	US20170127622A1	Smart Control IoT System for Agriculture Environment Control	USPTO	10 Nov 2016	Medium
5	US20130308060A1	Smart Container for Monitoring a Growing Plant	USPTO	5 Dec 2013	Low
6	US11632914B2	Smart Plant Cultivation Device & System Using IoT	USPTO	2024	Medium

Detailed Analysis — Patent 1

Patent ID: EP3990913A1

Title: Automated Plant Monitoring Systems and Methods
Office: European Patent Office
Publication Date: 4 May 2022

Technical Summary:

This patent describes an automated plant monitoring system using multiple environmental sensors. The system collects data, processes it using a controller, and provides plant health assessment and automated responses.

Key Independent Claim :

A system comprising:

- Environmental sensors
- Processing unit
- Automated plant care response mechanism
- Communication module

Mapping to Our Project :

Our project includes:

- Soil moisture sensor
- Temperature and humidity sensor
- Microcontroller (NodeMCU/Arduino)
- Automated irrigation

Design-around / Recommendations:

- Include AI-based recommendation layer
- Use solar-powered module
- Focus on small-scale implementation

Risk Assessment :

Medium Risk — because both systems involve sensor-based automated plant care.

Detailed Analysis — Patent 2

Patent ID: US20240329614A1

Title: Smart Plant Management System
Office: USPTO
Publication Date: 3 October 2024

Technical Summary

Describes a cloud-connected smart plant management system integrating IoT sensors, remote monitoring, and automated decision-making.

Key Independent Claim :

IoT-enabled plant management platform with cloud analytics and remote mobile application.

Mapping to Our Project :

Our system also includes cloud monitoring and mobile dashboard.

Design-around / Recommendations :

- Add edge processing before cloud upload
- Focus on educational/home gardening

Risk Assessment:

Low Risk — claims are broad and pending.

Detailed Analysis — Patent 3

Patent ID: US10349584B2

Title: System and Method for Plant Monitoring
Office: USPTO
Grant Date: 16 July 2019

Technical Summary

Describes monitoring plant health using environmental and soil data, processed by a controller to determine watering schedule.

Key Independent Claim

System for monitoring plant parameters and determining irrigation timing.

Mapping to Our Project :

Similar to our automated irrigation logic.

Design-around / Recommendations:

- Use multi-sensor weighted algorithm
- Include adaptive irrigation scheduling

Risk Assessment :

Medium Risk — watering logic similarity.

Detailed Analysis — Patent 4

Patent ID: US20170127622A1

Title: Smart Control IoT System for Agriculture Environment Control
Office: USPTO
Publication Date: 10 Nov 2016

Technical Summary:

Describes IoT-based environmental monitoring for agricultural fields including humidity, temperature, and light sensors.

Key Independent Claim :

IoT network controlling agricultural environmental parameters.

Mapping to Our Project :

Our project monitors environment but at smaller scale.

Design-around / Recommendations:

- Limit to home gardening
- Avoid large-scale distributed IoT network

Risk Assessment:

Medium Risk for large-scale agricultural deployment.

Detailed Analysis — Patent 5

Patent ID: US20130308060A1

Title: Smart Container for Monitoring a Growing Plant
Office: USPTO
Publication Date: 5 Dec 2013

Technical Summary:

Smart plant pot with embedded moisture sensor and monitoring display.

Key Independent Claim :

Container-integrated plant monitoring system.

Mapping to Our Project :

Our project is not container-integrated but separate IoT module.

Design-around / Recommendations:

Keep system external and modular.

Risk Assessment:

- Risk: Low

Detailed Analysis — Patent 6

Patent ID: US11632914B2

Title: Smart Plant Cultivation Device & System Using IoT
Office: USPTO
Grant Year: 2024

Technical Summary:

IoT-based plant cultivation system with automatic watering, lighting control, and remote monitoring.

Key Independent Claim :

Integrated IoT system for plant growth control and monitoring.

Mapping to Our Project :

Our project overlaps in IoT-based irrigation and monitoring.

Design-around / Recommendations:

- Add unique adaptive logic
- Avoid full automation of lighting system
- Differentiate by modular architecture

Risk Assessment:

Medium Risk.

4. Final Conclusion

The Smart Plant Monitoring System overlaps partially with existing patents, especially in automated irrigation and IoT monitoring. However, by introducing adaptive logic, hybrid edge-cloud architecture, and small-scale application focus, infringement risk can be reduced significantly.